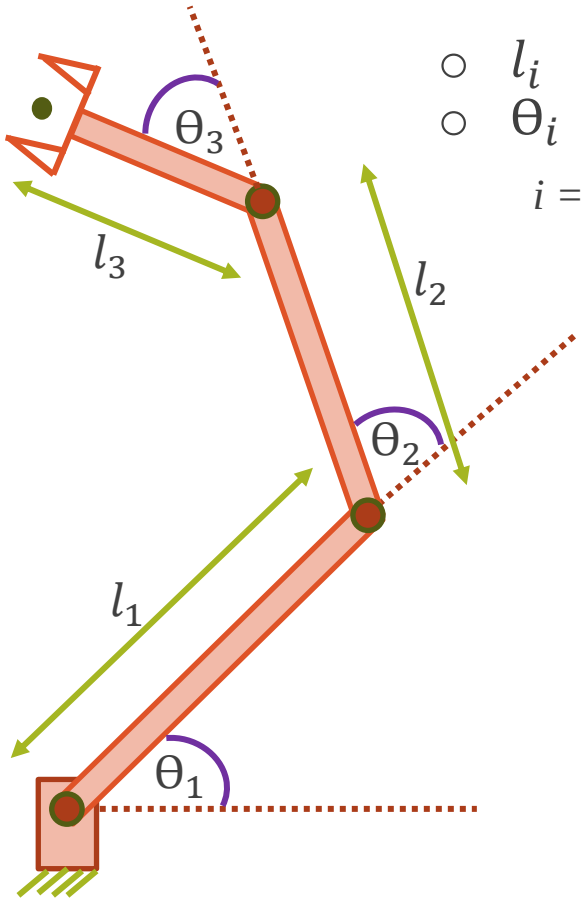
Kinematics of 3R or RRR Planar Robotic Arm

Presented by: Sourav Kumar

Forward Kinematics

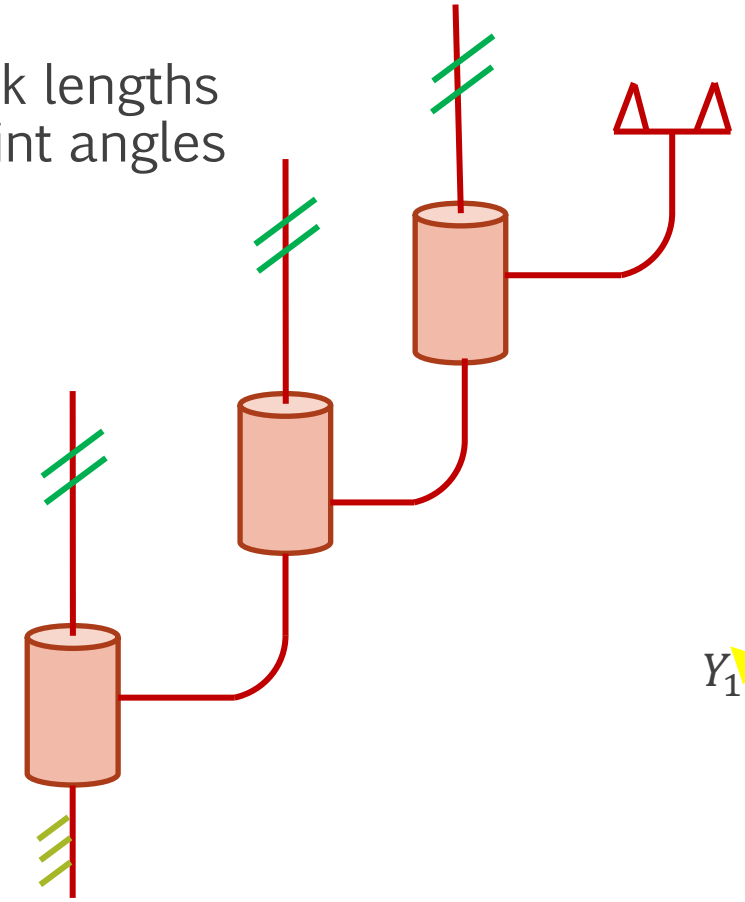
- All three joints are revolute.
  - Double hash marks indicates all 3 axes are parallel.



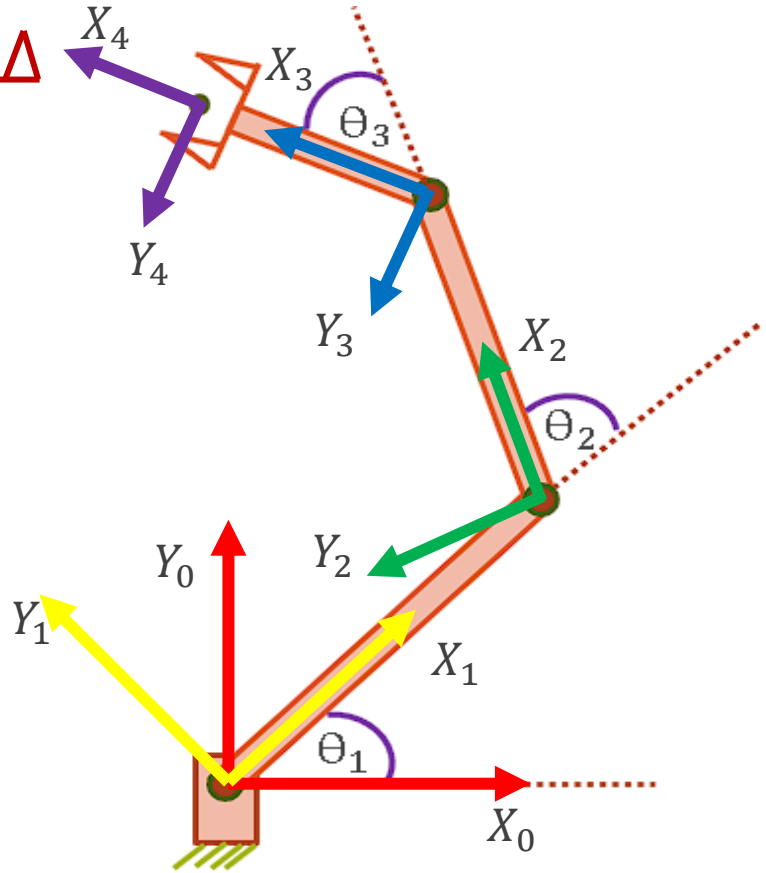
- l_i → Link lengths
- θ_i → Joint angles

$i = 1, 2, 3$

Kinematic diagram



Schematic Diagram

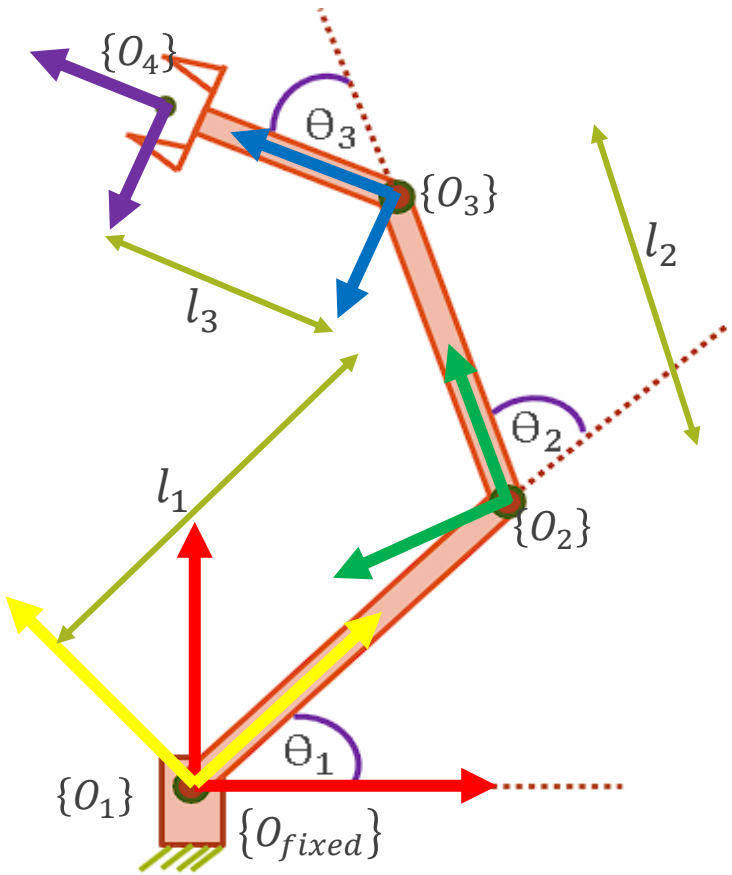


Assigning Frames

DH Parameters

$\boldsymbol{i}$	α_{i-1}	$\boldsymbol{a}_{i-1}$	$\boldsymbol{d}_i$	θ_i
1	0	0	0	θ_1
2	0	l_1	0	θ_2
3	0	l_2	0	θ_3
4	0	l_3	0	0

- $\{O_{fixed}\}\rightarrow$ Reference frame
- $\{O_1\}, \{O_2\}, \{O_3\} \rightarrow$ Intermediate frames
- $\{O_4\}\rightarrow$ Tip or End-Effector frame
- $\alpha_{i-1}\rightarrow$ Link twists
- $\boldsymbol{a}_{i-1}\rightarrow$ Link lengths
- $\boldsymbol{d}_i \rightarrow$ Link offsets
- $\theta_i \rightarrow$ Joint angles



Link Frame Assignment

Transformation of frames

- ${}^{i-1}_{i}[T]$ = Screw$_X(\boldsymbol{a_{i-1}}, \alpha_{\boldsymbol{i-1}})$ Screw$_Z(\boldsymbol{d_i}, \theta_{\boldsymbol{i}})$

$$= \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1} \\ \sin\theta_i \cos\alpha_{i-1} & \cos\theta_i \cos\alpha_{i-1} & -\sin\alpha_{i-1} & -\sin\alpha_{i-1} d_i \\ \sin\theta_i \sin\alpha_{i-1} & \cos\theta_i \sin\alpha_{i-1} & \cos\alpha_{i-1} & \cos\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ${}^{0}_{4}[T]$ = ${}^{0}_{1}[T]$ ${}^{1}_{2}[T]$ ${}^{2}_{3}[T]$ ${}^{3}_{4}[T]$

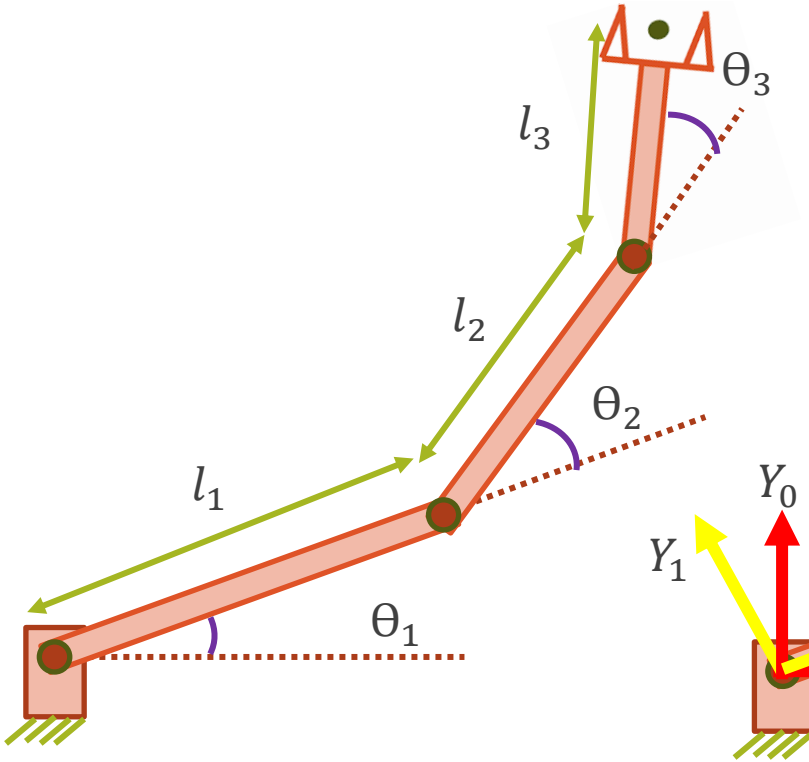
$$= \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & L_1 \\ \sin\theta_2 & \cos\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & L_2 \\ \sin\theta_3 & \cos\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \left[\begin{array}{c:c} {}^{0}_{4}[R]_{3X3} & {}^{0}_{4}\overrightarrow{p}_{3X1} \\ \hdashline 0_{3X1} & 1 \end{array}\right]_{4 X 4}$$

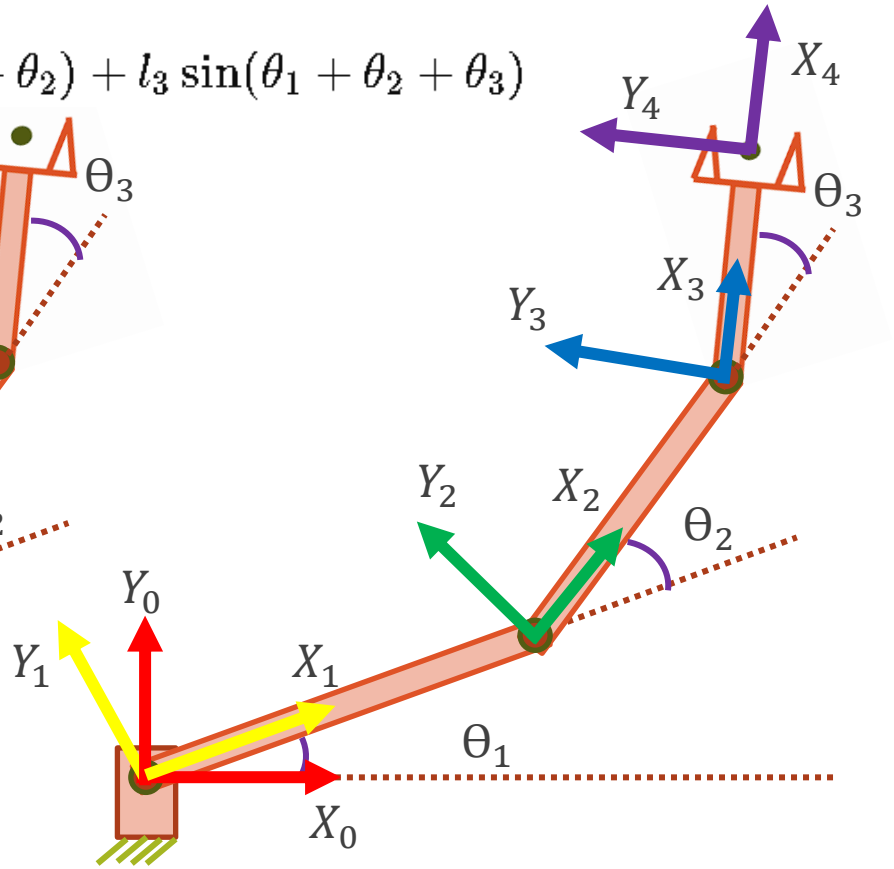
- ${}^{0}_{4}[R]_{3X3}$, represents orientation of tip w.r.t. fixed base
- ${}^{0}_{4}\overrightarrow{p}_{3X1}$, represents the position of tip/end effector w.r.t. fixed base

Inverse Kinematics

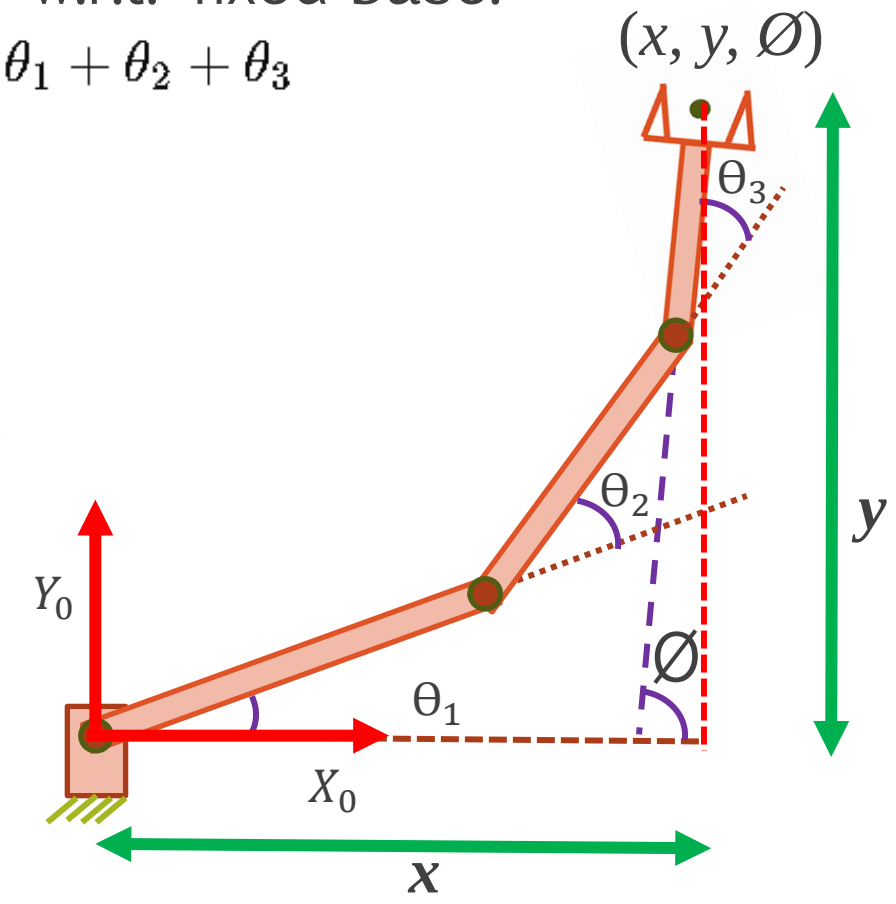
- Position and orientation of the of the end-effector w.r.t. fixed base.
  - $x = l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3)$
  - $y = l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)$
  - $\phi = \theta_1 + \theta_2 + \theta_3$



Kinematic diagram

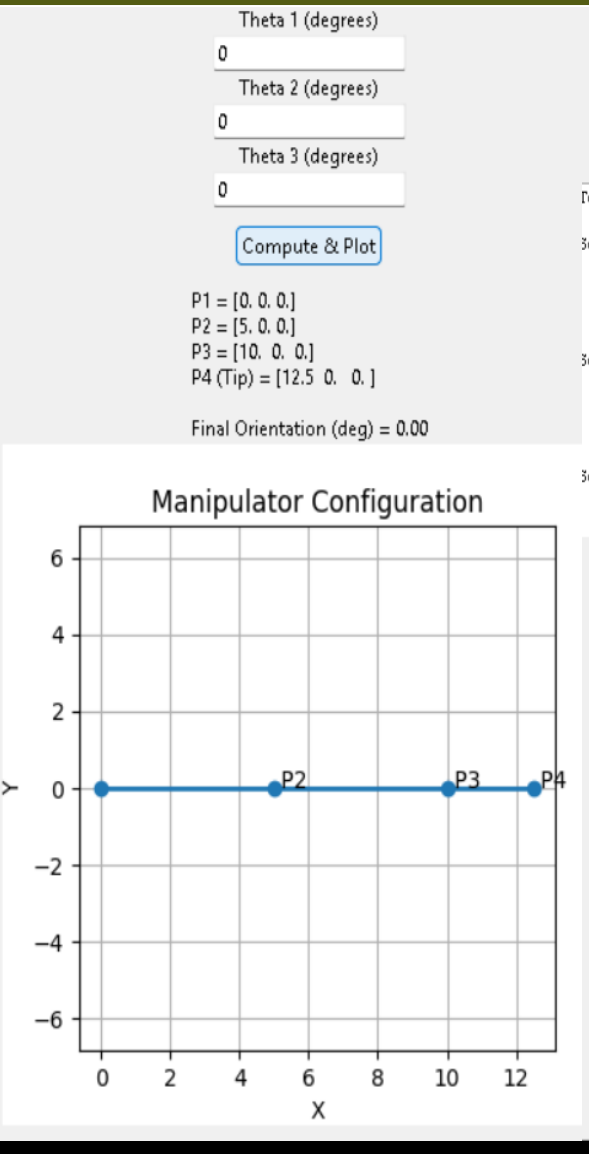


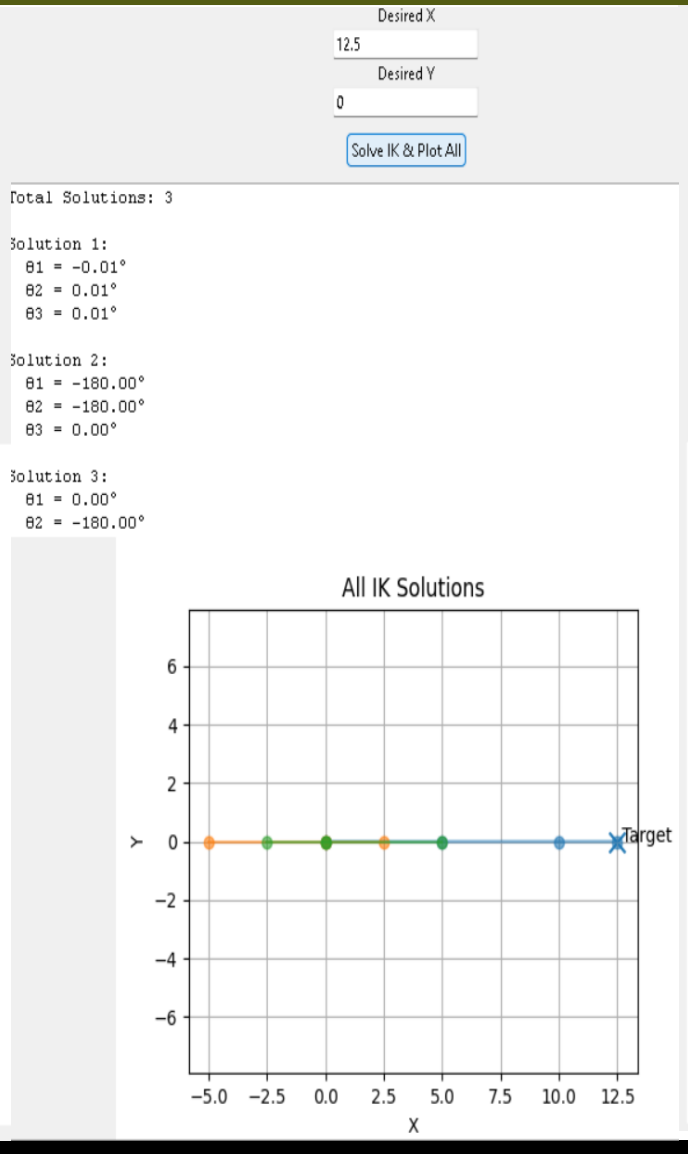
Link Frame Assignment

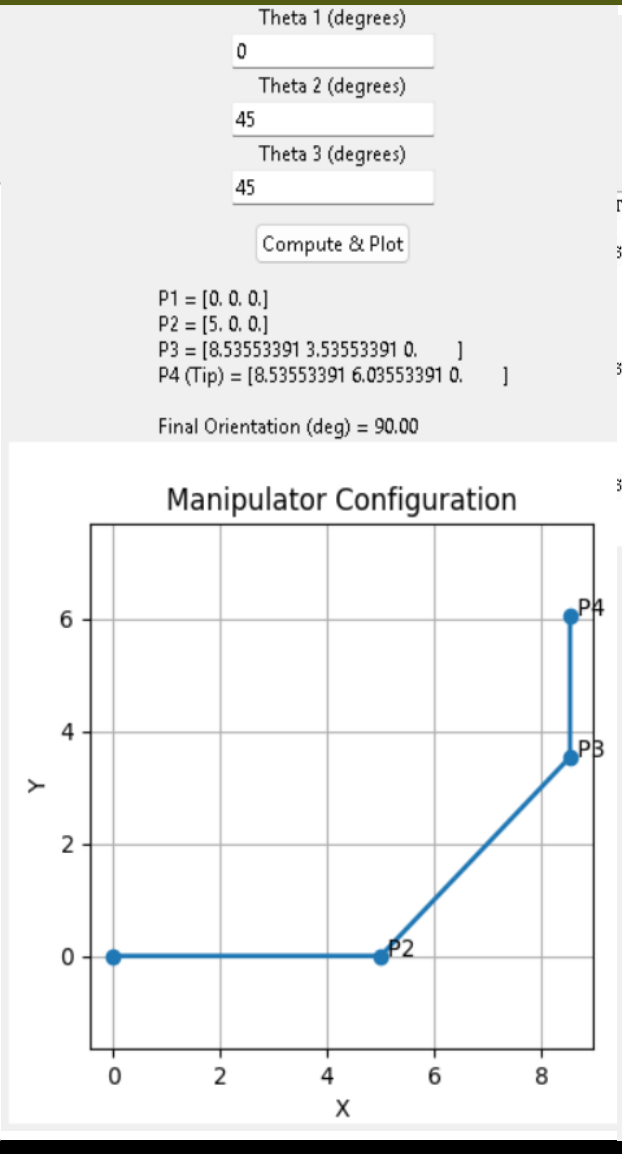


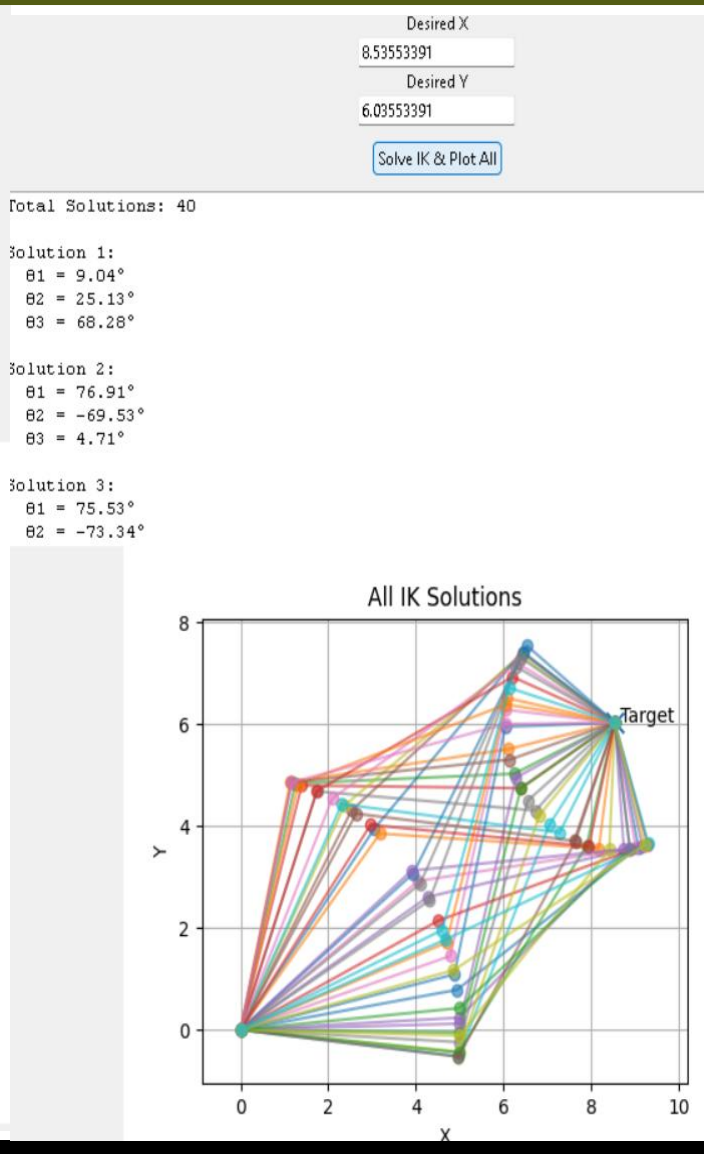
Projection of tip/end-effector's point

Results


Theta 1 (degrees)
0
Theta 2 (degrees)
0
Theta 3 (degrees)
0
Compute & Plot
P1 = [0. 0. 0.]
P2 = [5. 0. 0.]
P3 = [10. 0. 0.]
P4 (Tip) = [12.5 0. 0.]
Final Orientation (deg) = 0.00
Manipulator Configuration
P2
P3
P4
X
Y


Desired X
12.5
Desired Y
0
Solve IK & Plot All
Total Solutions: 3
Solution 1:
θ1 = -0.01°
θ2 = 0.01°
θ3 = 0.01°
Solution 2:
θ1 = -180.00°
θ2 = -180.00°
θ3 = 0.00°
Solution 3:
θ1 = 0.00°
θ2 = -180.00°
All IK Solutions
Target
X
Y


Theta 1 (degrees)
0
Theta 2 (degrees)
45
Theta 3 (degrees)
45
Compute & Plot
P1 = [0. 0. 0.]
P2 = [5. 0. 0.]
P3 = [8.53553391 3.53553391 0.]
P4 (Tip) = [8.53553391 6.03553391 0.]
Final Orientation (deg) = 90.00
Manipulator Configuration
P4
P3
P2
X
Y


Desired X
8.53553391
Desired Y
6.03553391
Solve IK & Plot All
Total Solutions: 40
Solution 1:
θ1 = 9.04°
θ2 = 25.13°
θ3 = 68.28°
Solution 2:
θ1 = 76.91°
θ2 = -69.53°
θ3 = 4.71°
Solution 3:
θ1 = 75.53°
θ2 = -73.34°
All IK Solutions
Target
X
Y

Results

Theta 1 (degrees)
45
Theta 2 (degrees)
45
Theta 3 (degrees)
45
Compute & Plot

P1 = [0. 0. 0.]
P2 = [3.53553391 3.53553391 0.]
P3 = [3.53553391 8.53553391 0.]
P4 (Tip) = [1.76776695 10.30330086 0.]

Final Orientation (deg) = 135.00

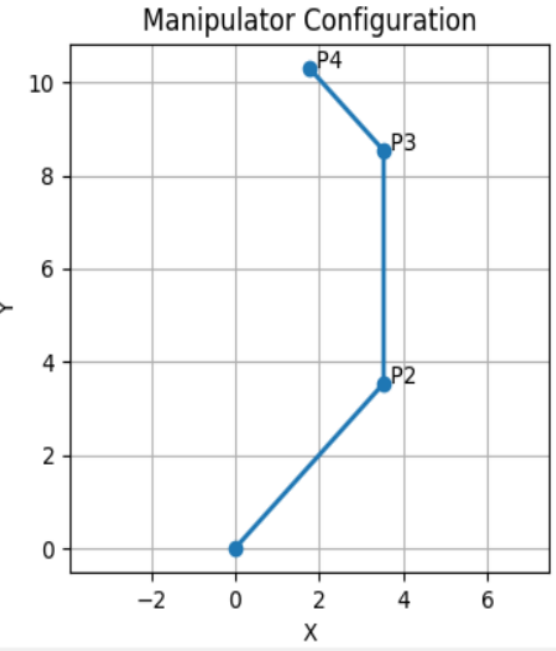
Desired X
1.76776695
Desired Y
10.30330086
Solve IK & Plot All

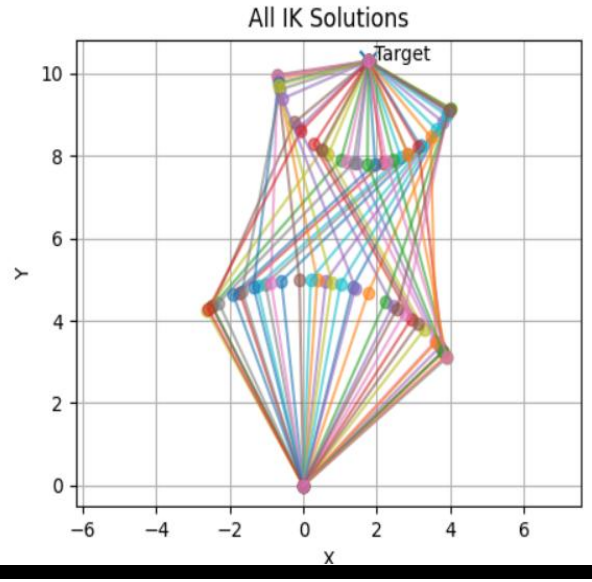
```
Total Solutions: 37

Solution 1:
  θ1 = 74.44°
  θ2 = -15.87°
  θ3 = 92.21°

Solution 2:
  θ1 = 69.34°
  θ2 = -5.76°
  θ3 = 89.11°

Solution 3:
  θ1 = 119.66°
  θ2 = -74.08°
```





Theta 1 (degrees)
20
Theta 2 (degrees)
30
Theta 3 (degrees)
60
Compute & Plot

P1 = [0. 0. 0.]
P2 = [4.6984631 1.71010072 0.]
P3 = [7.91240115 5.54032293 0.]
P4 (Tip) = [7.05735079 7.88955448 0.]

Final Orientation (deg) = 110.00

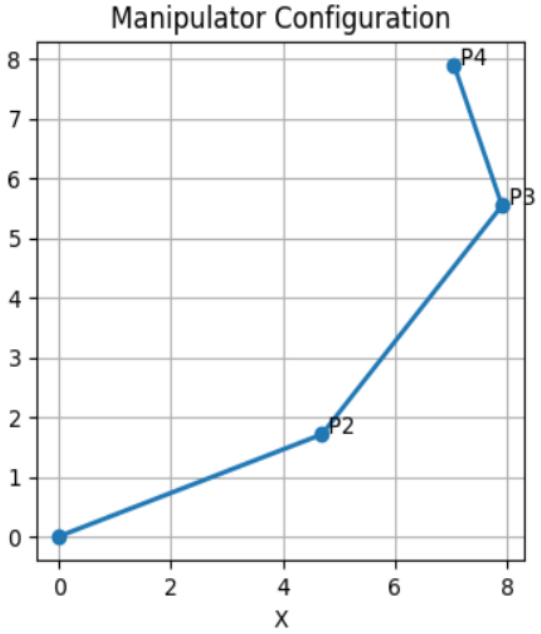
Desired X
7.05735079
Desired Y
7.88955448
Solve IK & Plot All

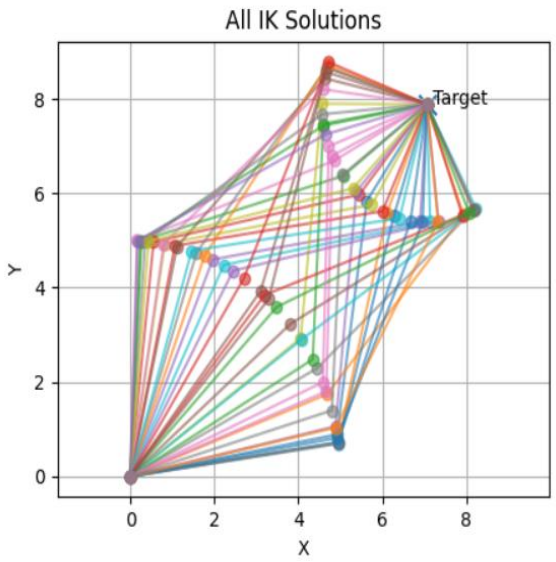
```
Total Solutions: 38

Solution 1:
  θ1 = 10.23°
  θ2 = 71.62°
  θ3 = -26.70°

Solution 2:
  θ1 = 20.39°
  θ2 = 29.16°
  θ3 = 60.89°

Solution 3:
  θ1 = 86.98°
  θ2 = -70.95°
```





Thank You